

SELBERG FEEDBACK AND APPROXIMATE DIVISOR DATA DO NOT STABLY DETERMINE SQUARE-ROOT PRIME ERROR

MAYK LOIDE BACCARO
INDEPENDENT RESEARCHER
M.BACCARO7663@STUDENT.NU.EDU

ABSTRACT. Selberg’s symmetry formula is strong enough to prove the prime number theorem. Approximate versions of its feedback can lose square-root-scale information about prime error. We construct positive weights on the actual prime powers that are constant along every prime-power chain and satisfy the Selberg symmetry formula with its classical $O(x)$ remainder, yet whose summatory functions separate by x^ρ for any prescribed $\rho \in (1/2, 1)$. After scale normalization, each complete divisor-residual vector through $2X$ has norm at most a fixed tolerance τ (and the two vectors are therefore at distance at most 2τ), while the summatory errors at X differ by $\gg_\rho X^\rho / \log X$. Thus every exact recovery map on this class has condition number at least $\gg_\rho X^\rho / \log X$ in the supremum norm. The same construction has centered divisor residual, bounded mean square, a compactly supported dyadic transform obstruction, and controlled generalized von Mangoldt recursions at every order. These results isolate a precise information barrier for the stated approximate summaries. Exact symbolic use of $\Lambda * \mathbf{1} = \log$ lies outside this approximation problem, and the results leave the Riemann hypothesis unchanged.

1. INTRODUCTION

Write

$$\psi(x) = \sum_{n \leq x} \Lambda(n), \quad (f * g)(n) = \sum_{d|n} f(d)g(n/d),$$

where $\Lambda(p^k) = \log p$ and $\Lambda(n) = 0$ off the prime powers. The identity

$$(1) \quad \Lambda * \mathbf{1} = \log$$

determines Λ exactly by Möbius inversion. Stable recovery from approximate divisor data is a separate question. This paper quantifies its conditioning inside a narrow arithmetic class that preserves the most conspicuous structure of Λ : nonnegativity, support on the actual prime powers, and constancy on each chain $p, p^2, p^3, \dots$.

The Riemann hypothesis is equivalent to a square-root-scale estimate for $\psi(x) - x$, with the usual logarithmic allowance; this formulation goes back to von Koch [9]. Selberg’s elementary proof of the prime number theorem derives the symmetry formula

$$(2) \quad \sum_{n \leq x} \Lambda(n) \log n + \sum_{mn \leq x} \Lambda(m)\Lambda(n) = 2x \log x + O(x)$$

and closes a qualitative feedback argument [8]. We test whether positivity, the $O(x)$ Selberg remainder, and small error in (1) force a much sharper summatory estimate. The constructed family gives a quantitative failure.

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For a weight a define

$$(3) \quad A_a(x) = \sum_{n \leq x} a(n), \quad E_a(x) = A_a(x) - x,$$

$$(4) \quad R_a(n) = (a * \mathbf{1})(n) - \log n,$$

$$(5) \quad S_a(x) = \sum_{n \leq x} a(n) \log n + \sum_{mn \leq x} a(m)a(n),$$

where the second sum in (5) is over ordered pairs. Our main theorem constructs, at every large scale X , two admissible weights whose residual vectors $(R_a(n))_{n \leq 2X}$ are uniformly close but whose outputs $E_a(X)$ are far apart.

Section 2 states the conditioning theorem and its square-root impossibility corollary. Sections 3–5 build the fixed two-sign family and prove its arithmetic and Selberg properties. Section 6 gives a compact dyadic transform consequence, and Section 7 treats the generalized von Mangoldt recursion. Appendix A records a simpler dense model that exposes the underlying cancellation without the prime-power constraints.

Related results fall into three groups. Beurling’s generalized primes show how prime-number theorems respond to changes in the underlying prime system [3]; here the ordinary primes remain fixed and only the weight changes. Quantitative remainder theorems for generalized primes require explicit analytic hypotheses [5]. Ivić introduced functions generalizing von Mangoldt’s function, and Calderón García-Zárate developed a Selberg-type asymptotic variant for such functions [7, 4]. Generalized von Mangoldt sums also encode criteria equivalent to the Riemann hypothesis [1, 2], while prime-sensitive Lindelöf formulations give another exact equivalence [6]. None of these results supplies the uniform inverse estimate tested here. Our theorem addresses conditioning of recovery from approximate arithmetic summaries on a fixed positive prime-power cone.

2. THE CONDITIONING OBSTRUCTION

Definition 2.1. Let $\mathcal{C}$ be the set of weights $a : \mathbb{N} \rightarrow [0, \infty)$ such that

$$\text{supp}(a) \subseteq \{p^k : p \text{ prime}, k \geq 1\} \quad \text{and} \quad a(p^k) = a(p) \quad (k \geq 1).$$

For $X \geq 2$, write

$$\mathcal{R}_X(a) = (R_a(n))_{1 \leq n \leq \lfloor 2X \rfloor}$$

and equip residual vectors with the supremum norm.

Theorem 2.2 (Near-linear conditioning lower bound). *Fix $\rho \in (1/2, 1)$. There are a function $\delta_\rho : \mathbb{N} \rightarrow \mathbb{R}$, a constant $C_\rho > 0$, and $X_\rho \geq 2$ with the following property. For every $X \geq X_\rho$ and $0 < \tau \leq 1$, the weights*

$$(6) \quad a_{X,\pm} = \Lambda \pm \eta_X \delta_\rho, \quad \eta_X = \frac{\tau}{C_\rho \log(2X)},$$

belong to $\mathcal{C}$ and satisfy

$$(7) \quad \|\mathcal{R}_X(a_{X,\pm})\|_\infty \leq \tau,$$

$$(8) \quad |E_{a_{X,+}}(X) - E_{a_{X,-}}(X)| \geq \frac{\tau}{\rho C_\rho} \frac{X^\rho}{\log(2X)}.$$

Each $a_{X,\pm}$ also obeys

$$(9) \quad S_{a_{X,\pm}}(x) = 2x \log x + O_\rho(x) \quad (x \geq 2),$$

with a constant independent of X and τ .

Consequently, if a map

$$\Phi_X : \mathcal{R}_X(\mathcal{C}) \longrightarrow \mathbb{R}$$

TABLE 1. Properties of the fixed and scale-normalized pairs. All constants may depend on ρ .

Property	Fixed pair	Normalized pair at X
Actual prime-power support	yes	yes
Strictly positive at every prime power	yes	yes
Constant on every p^k chain	yes	yes
Centered divisor mean	$O(x^\rho)$	$O(x^\rho/\log X)$
Divisor residual mean square	$O(x)$	$O(x/\log^2 X)$
Complete residual through $2X$	$O(\log X)$	at most τ
Selberg statistic	$2x \log x + O(x)$	uniform $2x \log x + O(x)$
Two-sign output separation at X	$\asymp X^\rho$	$\gg \tau X^\rho/\log X$

recovers $E_a(X)$ exactly for every $a \in \mathcal{C}$, then its Lipschitz constant in the supremum norm satisfies

$$(10) \quad \text{Lip}(\Phi_X) \geq \frac{1}{2\rho C_\rho} \frac{X^\rho}{\log(2X)} \quad (X \geq X_\rho).$$

In particular, for every $\delta \in (0, 1/2)$ one can take $\rho = 1 - \delta$ and obtain

$$\text{Lip}(\Phi_X) \gg_\delta \frac{X^{1-\delta}}{\log X}.$$

The scale dependence in (6) is essential and explicit: the theorem measures the condition number of a family of finite-scale inverse problems. The fixed perturbations from which the family is obtained appear in Theorem 3.2.

Corollary 2.3 (No uniform square-root correction). *Fix $\varepsilon \in [0, 1/2)$. There is no constant C_ε such that every $a \in \mathcal{C}$ and every $X \geq 2$ satisfy*

$$(11) \quad |A_a(X) - \psi(X)| \leq C_\varepsilon X^{1/2+\varepsilon} \max_{n \leq 2X} |R_a(n)|.$$

The conclusion remains true after restricting to weights that satisfy (9) with a uniform $O_\rho(x)$ constant.

Proof. Choose $\rho \in (1/2 + \varepsilon, 1)$ and use one of the weights in (6). Since

$$A_{a_{X,\pm}}(X) - \psi(X) = \pm \eta_X \sum_{n \leq X} \delta_\rho(n),$$

the asymptotic established in Proposition 3.1 gives, for both signs and all sufficiently large X ,

$$|A_{a_{X,\pm}}(X) - \psi(X)| \geq \frac{\tau}{2\rho C_\rho} \frac{X^\rho}{\log(2X)}.$$

The right side of (11) is at most $C_\varepsilon \tau X^{1/2+\varepsilon}$. Their ratio tends to infinity. $\square$

3. A CENTERED PERTURBATION ON THE ACTUAL PRIME POWERS

Fix $\rho \in (1/2, 1)$ for the remainder of Sections 3–6, and define

$$(12) \quad c_2 = - \sum_{\substack{p \text{ prime} \\ p \text{ odd}}} \frac{p^{\rho-1} \log p}{p-1}, \quad c_p = p^{\rho-1} \log p \quad (p \text{ odd}).$$

The series defining c_2 converges because its summand has order $\log p/p^{2-\rho}$ and $2 - \rho > 1$. Extend the coefficients to an arithmetic function by

$$(13) \quad \delta_\rho(p^k) = c_p \quad (k \geq 1), \quad \delta_\rho(n) = 0 \quad (n \text{ not a prime power}).$$

The correction on the 2-power chain enforces the exact centering identity

$$(14) \quad \sum_p \frac{c_p}{p-1} = 0.$$

Set

$$(15) \quad C_\rho = \max \left\{ 3^{\rho-1}, \frac{|c_2|}{\log 2} \right\}, \quad D_\rho = \sum_{n \geq 1} \frac{|\delta_\rho(n)|}{n} = \sum_p \frac{|c_p|}{p-1}.$$

For odd primes, $|c_p|/\log p = p^{\rho-1} \leq 3^{\rho-1}$, so $|c_p| \leq C_\rho \log p$ for every prime p . Both constants are finite, and $C_\rho > 0$.

Proposition 3.1 (Size of the perturbation). *As $x \rightarrow \infty$,*

$$(16) \quad \sum_{n \leq x} |\delta_\rho(n)| = O_\rho(x^\rho),$$

$$(17) \quad H_\rho(x) := \sum_{n \leq x} \delta_\rho(n) = \frac{x^\rho}{\rho} + o_\rho(x^\rho).$$

Proof. Chebyshev's estimate $\vartheta(y) = \sum_{p \leq y} \log p = O(y)$ and partial summation give

$$\sum_{p \leq y} p^{\rho-1} \log p = O_\rho(y^\rho).$$

The levels p^k with $k \geq 2$ contribute $O_\rho(x^{\rho/2} \log x)$, and the correction along the powers of 2 contributes $O_\rho(\log x)$. This proves (16).

For the signed sum, the prime level contributes

$$\sum_{p \leq x} p^{\rho-1} \log p = \frac{x^\rho}{\rho} + o_\rho(x^\rho)$$

by the prime number theorem and partial summation. The higher prime-power levels and the 2-chain correction are $o_\rho(x^\rho)$ by the bounds just given. Hence (17) follows. $\square$

Theorem 3.2 (Fixed two-sign family). *For every*

$$(18) \quad 0 < \eta \leq \min\{1, (2C_\rho)^{-1}\},$$

the weights

$$a_\pm = \Lambda \pm \eta \delta_\rho$$

belong to $\mathcal{C}$, are strictly positive on every prime power, and satisfy

$$(19) \quad A_{a_+}(x) - A_{a_-}(x) = \frac{2\eta}{\rho} x^\rho + o_\rho(x^\rho).$$

Proof. Support and chain constancy follow from (13). At p^k ,

$$a_\pm(p^k) = \log p \pm \eta c_p \geq (1 - \eta C_\rho) \log p \geq \frac{1}{2} \log p > 0.$$

Equation (19) is twice η times (17). $\square$

4. THE DIVISOR RESIDUAL

Let

$$(20) \quad r_\rho = \delta_\rho * \mathbf{1}.$$

Unique factorization turns the convolution into the explicit additive function

$$(21) \quad r_\rho(n) = \sum_p c_p v_p(n),$$

where $v_p(n)$ is the exponent of p in n . Thus

$$(22) \quad |r_\rho(n)| \leq C_\rho \sum_p v_p(n) \log p = C_\rho \log n.$$

Proposition 4.1 (Centered mean and bounded mean square). *For $x \geq 2$,*

$$(23) \quad \sum_{n \leq x} r_\rho(n) = O_\rho(x^\rho),$$

$$(24) \quad \sum_{n \leq x} |r_\rho(n)|^2 = O_\rho(x).$$

Consequently, the residuals of the fixed weights satisfy

$$R_{a_\pm}(n) = \pm \eta r_\rho(n)$$

with centered mean, bounded mean square, and the pointwise estimate $|R_{a_\pm}(n)| \leq \eta C_\rho \log n$.

Proof. For each prime $p \leq x$,

$$(25) \quad \sum_{n \leq x} v_p(n) = \sum_{\substack{k \geq 1 \\ p^k \leq x}} \left\lfloor \frac{x}{p^k} \right\rfloor = \frac{x}{p-1} + O\left(1 + \frac{\log x}{\log p}\right).$$

Insert (25) into (21). The complete main term vanishes by (14). Partial summation with $\vartheta(y) = O(y)$ bounds the omitted tail by

$$x \sum_{p > x} \frac{|c_p|}{p-1} = O_\rho(x^\rho),$$

while

$$\sum_{p \leq x} |c_p| \left(1 + \frac{\log x}{\log p}\right) = O_\rho(x^\rho).$$

This proves (23).

For the second moment, use

$$(26) \quad \sum_{n \leq x} v_p(n)^2 \leq x \sum_{k, l \geq 1} p^{-\max(k, l)} = x \frac{p+1}{(p-1)^2},$$

$$(27) \quad \sum_{n \leq x} v_p(n) v_q(n) \leq x \sum_{k, l \geq 1} p^{-k} q^{-l} = \frac{x}{(p-1)(q-1)} \quad (p \neq q).$$

Expanding the square in (21), taking absolute values on the cross terms, and applying (26)–(27) gives

$$\sum_{n \leq x} |r_\rho(n)|^2 \leq x \sum_p |c_p|^2 \frac{p+1}{(p-1)^2} + 2x \sum_{p < q} \frac{|c_p c_q|}{(p-1)(q-1)}.$$

The first series converges because its general term is $O_\rho((\log p)^2/p^{3-2\rho})$; the second is bounded by $x D_\rho^2$. $\square$

5. SELBERG FEEDBACK SURVIVES THE PERTURBATION

Proposition 5.1 (Uniform Selberg remainder). *Let $a = \Lambda + e$, where $e = \pm \eta \delta_\rho$ and $|\eta| \leq \eta_0$ for a fixed η_0 . Then*

$$(28) \quad S_a(x) = 2x \log x + O_{\rho, \eta_0}(x).$$

Proof. Expanding the ordered convolution gives

$$(29) \quad S_a(x) - S_\Lambda(x) = \sum_{n \leq x} e(n) \log n + 2 \sum_{mn \leq x} e(m) \Lambda(n) + \sum_{mn \leq x} e(m) e(n).$$

The first term is $O_{\rho, \eta_0}(x^\rho \log x) = O_{\rho, \eta_0}(x)$ by (16). Choose C_ψ so that $\psi(y) \leq C_\psi y$ for $y \geq 1$. Then

$$\sum_{mn \leq x} |e(m)| \Lambda(n) \leq C_\psi x \sum_{m \geq 1} \frac{|e(m)|}{m} \leq C_\psi \eta_0 D_\rho x.$$

Also

$$\sum_{mn \leq x} |e(m) e(n)| \leq x \left(\sum_{m \geq 1} \frac{|e(m)|}{m} \right)^2 \leq \eta_0^2 D_\rho^2 x.$$

Combine these estimates with Selberg's formula (2). □

Proof of Theorem 2.2. Use the perturbation (13). Since $X \geq 2$ and $\tau \leq 1$,

$$\eta_X C_\rho = \frac{\tau}{\log(2X)} \leq \frac{1}{\log 4} < 1.$$

The positivity argument from Theorem 3.2 therefore applies globally to $a_{X, \pm}$. Equations (4), (1), and (22) yield, for $n \leq 2X$,

$$|R_{a_{X, \pm}}(n)| = \eta_X |r_\rho(n)| \leq \eta_X C_\rho \log(2X) = \tau.$$

By (17), enlarge X_ρ until $|H_\rho(X)| \geq X^\rho / (2\rho)$. Then

$$|E_{a_{X, +}}(X) - E_{a_{X, -}}(X)| = 2\eta_X |H_\rho(X)| \geq \frac{\tau}{\rho C_\rho} \frac{X^\rho}{\log(2X)}.$$

Proposition 5.1 applies uniformly because $\eta_X \leq (C_\rho \log 4)^{-1}$. Finally,

$$\|\mathcal{R}_X(a_{X, +}) - \mathcal{R}_X(a_{X, -})\|_\infty \leq 2\tau.$$

Taking the quotient of the output separation by this input distance proves (10). □

6. A COMPACT DYADIC TRANSFORM

The full residual vector in Theorem 2.2 is not the only summary that loses the x^ρ direction. A simple dyadic combination produces an exactly compactly supported kernel.

Define

$$(30) \quad K(u) = \begin{cases} 0, & u \leq 0, \\ \frac{1}{2}, & 0 < u \leq 1, \\ \frac{1}{2u^2}, & u > 1, \end{cases}$$

and set

$$(31) \quad T_a(X) = \sum_{n \geq 1} a(n) K(n/X), \quad M(X) = \int_1^\infty K(t/X) dt, \quad \mathcal{E}_a(X) = M(X) - T_a(X).$$

The series converges absolutely for the weights considered here.

Proposition 6.1 (Compact dyadic separation). *For the fixed pair $a_\pm = \Lambda \pm \eta \delta_\rho$, define*

$$\mathcal{D}_a(X) = 4\mathcal{E}_a(X) - \mathcal{E}_a(2X).$$

Then

$$(32) \quad \mathcal{D}_{a_+}(X) - \mathcal{D}_{a_-}(X) = -2\eta(4 - 2^\rho) I_\rho X^\rho + o_\rho(X^\rho), \quad I_\rho = \frac{1}{\rho(2 - \rho)}.$$

Equivalently, the perturbation is tested against the compactly supported kernel

$$(33) \quad W(u) = 4K(u) - K(u/2) = \begin{cases} \frac{3}{2}, & 0 < u \leq 1, \\ \frac{2}{u^2} - \frac{1}{2}, & 1 < u \leq 2, \\ 0, & u > 2. \end{cases}$$

If

$$\mathcal{Q}_a(Y) = \sum_{\substack{j \geq 0 \\ 2^j \leq Y}} |\mathcal{D}_a(2^j)|^2,$$

then for every $\varepsilon < 2\rho - 1$, at least one of a_+ and a_- fails the estimate $\mathcal{Q}_a(Y) = O(Y^{1+\varepsilon})$.

Proof. Stieltjes summation and (17) give

$$(34) \quad \sum_{n \geq 1} \delta_\rho(n) K(n/X) = I_\rho X^\rho + o_\rho(X^\rho),$$

where

$$I_\rho = \int_0^\infty u^{\rho-1} K(u) du = \frac{1}{2\rho} + \frac{1}{2(2-\rho)} = \frac{1}{\rho(2-\rho)}.$$

Since the common main term M cancels between signs,

$$\mathcal{D}_{a_+}(X) - \mathcal{D}_{a_-}(X) = -2\eta \sum_{n \geq 1} \delta_\rho(n) W(n/X).$$

Equations (34) and (33) yield (32).

For all sufficiently large dyadic X ,

$$|\mathcal{D}_{a_+}(X)|^2 + |\mathcal{D}_{a_-}(X)|^2 \geq \frac{1}{2} |\mathcal{D}_{a_+}(X) - \mathcal{D}_{a_-}(X)|^2 \gg_{\rho, \eta} X^{2\rho}.$$

Summing over dyadic $X \leq Y$ gives $\mathcal{Q}_{a_+}(Y) + \mathcal{Q}_{a_-}(Y) \gg_{\rho, \eta} Y^{2\rho}$. At least one fixed sign therefore violates every $O(Y^{1+\varepsilon})$ bound with $1 + \varepsilon < 2\rho$. $\square$

7. GENERALIZED VON MANGOLDT RECURSIONS

Following the generalized von Mangoldt hierarchy studied by Ivić and in subsequent Selberg-type variants [7, 4, 2], define the functions by

$$(35) \quad \Lambda_k = \mu * \log^k.$$

They are nonnegative and satisfy the recursion

$$(36) \quad \Lambda_{k+1} = \Lambda_k \log + \Lambda_k * \Lambda \quad (k \geq 1).$$

For a positive weight a , define its companion hierarchy by

$$(37) \quad G_{a,1} = a, \quad G_{a,k+1} = G_{a,k} \log + G_{a,k} * a.$$

This recursion agrees with (36) at $a = \Lambda$ and preserves nonnegativity.

Lemma 7.1 (Weighted prefix bound). *Let $f, g \geq 0$ satisfy*

$$\sum_{n \leq y} f(n) \leq Qy \log^m(2y), \quad \sum_{n \leq y} g(n) \leq Cy \quad (y \geq 1).$$

With $q = 1 + 1/\log 2$,

$$\sum_{n \leq x} (f * g)(n) \leq qCQx \log^{m+1}(2x).$$

Proof. Partial summation gives

$$\sum_{n \leq x} \frac{g(n)}{n} \leq C(1 + \log x) \leq qC \log(2x).$$

Therefore

$$\sum_{uv \leq x} f(u)g(v) \leq Qx \log^m(2x) \sum_{v \leq x} \frac{g(v)}{v},$$

which is the asserted estimate. $\square$

Theorem 7.2 (Order-uniform hierarchy control). *Let $a = a_+$ or a_- be a fixed weight from Theorem 3.2. There exist constants $B, C \geq 1$, depending on ρ and η but not on k or x , such that*

$$(38) \quad \sum_{n \leq x} |G_{a,k}(n) - \Lambda_k(n)| \leq BC^{k-2}x \log^{k-2}(2x) \quad (k \geq 2, x \geq 1).$$

Proof. Write $e = a - \Lambda$ and $\Delta_k = G_{a,k} - \Lambda_k$. Choose constants C_U, C_ψ such that

$$\sum_{n \leq x} |e(n)| \leq C_U x^\rho, \quad \psi(x) \leq C_\psi x \quad (x \geq 1),$$

and retain $D = \sum |e(n)|/n < \infty$. The expansion at order two is

$$\Delta_2 = e \log + e * \Lambda + \Lambda * e + e * e.$$

Since $x^{\rho-1} \log x \leq [e(1-\rho)]^{-1}$ for $x \geq 1$, the argument of Proposition 5.1 gives

$$(39) \quad \sum_{n \leq x} |\Delta_2(n)| \leq B_0 x, \quad B_0 = \frac{C_U}{e(1-\rho)} + 2C_\psi D + D^2.$$

Set

$$q = 1 + \frac{1}{\log 2}, \quad C_\Lambda = 1 + qC_\psi, \quad C_a = C_\psi + C_U.$$

Induction from (36) and Lemma 7.1 yields

$$(40) \quad \sum_{n \leq x} \Lambda_k(n) \leq C_\psi C_\Lambda^{k-1} x \log^{k-1}(2x).$$

Subtracting (36) from (37) gives

$$(41) \quad \Delta_{k+1} = \Delta_k \log + \Delta_k * a + \Lambda_k * e.$$

Take

$$B \geq \max\{1, B_0, C_\psi D\}, \quad C \geq 2 \max\{1 + qC_a, C_\Lambda\}.$$

Assume (38) at order k . The first two terms in (41), using Lemma 7.1 for the convolution, contribute at most

$$BC^{k-2}(1 + qC_a)x \log^{k-1}(2x) \leq \frac{1}{2}BC^{k-1}x \log^{k-1}(2x).$$

Using (40) and $\sum |e(n)|/n = D$, the last term contributes at most

$$C_\psi DC_\Lambda^{k-1}x \log^{k-1}(2x) \leq \frac{1}{2}BC^{k-1}x \log^{k-1}(2x).$$

This closes the induction. $\square$

Theorem 7.2 is deliberately order-uniform: it prints the full C^{k-2} dependence, with all growth in k visible. For this constructed pair, every displayed companion recursion obeys the stated order-uniform envelope while the original x^ρ summatory separation persists. These bounds alone therefore do not exclude the pair.

8. CONCLUDING PERSPECTIVE

The construction separates four statements that can look deceptively close.

- (i) The exact convolution equation $a * \mathbf{1} = \log$ identifies $a = \Lambda$.
- (ii) A small residual $a * \mathbf{1} - \log$ permits unstable recovery of $A_a(X) - x$.
- (iii) Positivity, actual prime-power support, chain constancy, centered first and second residual moments, and an $O(x)$ Selberg remainder still permit poor square-root conditioning.
- (iv) Higher generalized von Mangoldt recursions stay quantitatively controlled along the same unstable family.

Exact algebraic uniqueness coexists with an inverse-problem condition number that grows almost linearly with the observation scale. Any successful square-root recovery argument from approximate divisor or Selberg data must introduce a norm, cancellation principle, spectral constraint, or analytic input that detects the signed prime-level direction constructed here.

APPENDIX A. A DENSE WARM-UP

The following model removes the prime-power restrictions and displays the same Selberg cancellation in one line of construction.

Proposition A.1. *Fix $c > 0$ and $\rho \in (1/2, 1)$, and set*

$$a(n) = 1 + c\rho n^{\rho-1}.$$

Then

$$(42) \quad \sum_{n \leq x} a(n) = x + cx^\rho + O_{c,\rho}(1),$$

$$(43) \quad S_a(x) = 2x \log x + O_{c,\rho}(x).$$

Proof. Euler summation gives $\sum_{n \leq x} n^{\rho-1} = x^\rho/\rho + O_\rho(1)$, proving (42). For the constant weight 1,

$$\sum_{n \leq x} \log n + \sum_{mn \leq x} 1 = 2x \log x + O(x).$$

If $h(n) = c\rho n^{\rho-1}$, then

$$\sum_{n \leq x} h(n) \log n = O(x^\rho \log x) = O(x),$$

and

$$\sum_{mn \leq x} h(m) \leq x \sum_{m \geq 1} \frac{h(m)}{m} = O(x).$$

The same convergent weighted sum bounds the quadratic term $\sum_{mn \leq x} h(m)h(n)$ by $O(x)$. Expanding S_{1+h} proves the result. $\square$

APPENDIX B. VERIFICATION REMARKS

The construction uses no numerical fitting. The definition of c_2 makes the centering identity (14) immediate once the defining series and D_ρ are known to converge. In Proposition 3.1, only the prime level supplies the term x^ρ/ρ ; the higher prime powers and the corrected 2-chain are lower order. The residual calculation (21) uses the diagonal density $(p+1)/(p-1)^2$ and the off-diagonal density $1/((p-1)(q-1))$, while (29) uses the factor 2 from the ordered cross term.

For the compact transform, the untruncated Mellin coefficient is $\int_0^\infty u^{\rho-1} K(u) du = 1/(\rho(2-\rho))$. The dyadic combination $4K(u) - K(u/2)$ has compact support; K itself has an unbounded tail. The constants B and C^{k-2} in (38) are kept explicit because separate bounds for fixed k would not yield order-uniform control.

DATA AND CODE AVAILABILITY

The construction and every asymptotic estimate used in the theorems are contained in the manuscript; no external data or numerical experiment is required. An accompanying Lean 4 development checks the finite conditioning algebra: the distinguished correction, positivity of the signed perturbations, residual scaling, two-sign separation, and the resulting Lipschitz lower bound. The analytic construction, asymptotic estimates, and order-uniform hierarchy remain manuscript-level proofs.

ASSISTANCE DISCLOSURE

AI systems were used extensively in mathematical derivation, Lean proof development, literature discovery, organization, and typesetting. The author remains responsible for every mathematical statement, proof, citation, and submission decision.

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